ANSWER KEY

Lab 6 HTTP plain text credentials

Capture the traffic in Wireshark (Windows 11 PC, sign in with the password networkwalks) and follow the solution steps below.

Task 1. Capture the traffic — 10 marks

Sign in to the Windows PC and capture the network traffic. How many packets did Wireshark capture?

Answer: 255

Solution: Sign in with the password networkwalks, double-click Wireshark on the desktop, then double-click the WiFi interface. The status bar shows Packets: 255.

Task 2. Which protocol — 10 marks

A staff member signed in to the old staff portal. Which protocol carried the sign-in?

Answer: HTTP

Solution: Filter

```
http
```

. The sign-in packets say POST /login HTTP/1.1. The protocol is HTTP, which is not encrypted.

Task 3. The method — 10 marks

A web form sends the typed values to the server. Which HTTP method did the sign-in use?

Answer: POST

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The method is POST.

Task 4. The login page — 10 marks

Which page path did the form send the sign-in to?

Answer: /login

Solution: Filter

```
http.request.method == "POST"
```

. The Info column shows POST /login. The path is /login.

Task 5. The PC that signed in — 10 marks

Which PC sent the sign-in?

Answer: 192.168.10.45

Solution: Filter

```
http.request.method == "POST"
```

. The Source column shows 192.168.10.45.

Task 6. How many attempts — 10 marks

How many times did the user send the sign-in form?

Answer: 2

Solution: Filter

```
http.request.method == "POST"
```

. Two packets stay in the list. The user tried 2 times.

Task 7. The username — 10 marks

Click one POST packet. In the packet details, open HTML Form URL Encoded. What is the username?

Answer: farhan.k

Solution: Click a POST packet and open HTML Form URL Encoded. The line

```
username = farhan.k
```

shows the user.

Task 8. The password — 10 marks

The second sign-in worked (the page said “Welcome back”). What password did the user type that time?

Answer: Mango#2026

Solution: Click the second POST packet and open HTML Form URL Encoded. It shows

```
password = Mango#2026
```

. The first try (Mango2025) failed.

Task 9. Packet details line — 10 marks

Which line in the packet details shows the typed username and password in an easy list?

Answer: HTML Form URL Encoded

Solution: At the bottom of the packet details you find HTML Form URL Encoded. It lists the username and password.

Task 10. The fix — 10 marks

How can the company stop people from reading passwords like this?

Answer: Use HTTPS

Solution: HTTPS encrypts the traffic, so a sniffer sees only unreadable data. The answer is Use HTTPS.

SUMMARY

Protocol:	HTTP, POST /login
User and PC:	farhan.k from 192.168.10.45
Logins:	2 (1 failed, 1 worked)
Password:	Mango#2026
Fix:	Use HTTPS

Watch the Wireshark video playlist: https://www.youtube.com/playlist?list=PLG-wUGWU39ro